SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: CSA16 COURSE NAME: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2
Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:25**

Concept Mapping
Annexure A

Code of Conduct:

*I **J.Mohamed Aseem(192511258)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

J.Mohamed Aseem

Signature of the student:

RUBRICS FOR CALCULATION:

Concept Mapping					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Annexure A

Concept Map 1: Decision Support Systems (DSS)

Task Description:

Construct a concept map that explains how a Decision Support System (DSS) supports organizational decision-making.

What your map must show:

1. Core components: Data, Model, User Interface
2. Types of DSS
3. Flow: Data → Analysis → Decision
4. Role in business decision-making
5. Real-world example linkage

Concept Map 2: Operational Systems vs DSS

Task Description:

Develop a comparative concept map that clearly distinguishes Operational Systems (OLTP) and Decision Support Systems (OLAP).

What your map must show:

1. Definitions of OLTP and DSS
2. Differences (data type, purpose, users, processing)
3. Similarities and interactions
4. Real-time vs historical data
5. Example systems (banking vs analytics)

Concept Map 3: Architecture of Data Warehouse

Task Description:

Create a hierarchical concept map illustrating the architecture of a Data Warehouse.

What your map must show:

1. Data Sources → ETL → Data Warehouse → Data Marts
2. Metadata layer
3. Front-end tools (reporting, OLAP)
4. Data flow direction
5. Role of each layer

Concept Map 4: OLAP, Data Cubes & Dimensional Modeling

Task Description:

Design a concept map that connects OLAP operations, Data Cubes, and Dimensional Modeling techniques.

What your map must show:

1. Data cube structure (dimensions, measures)
2. OLAP operations (slice, dice, roll-up, drill-down)
3. Star schema and snowflake schema
4. Fact and dimension tables
5. Analytical usage

Concept Map 5: Query Processing & BI Tools (KNIME)

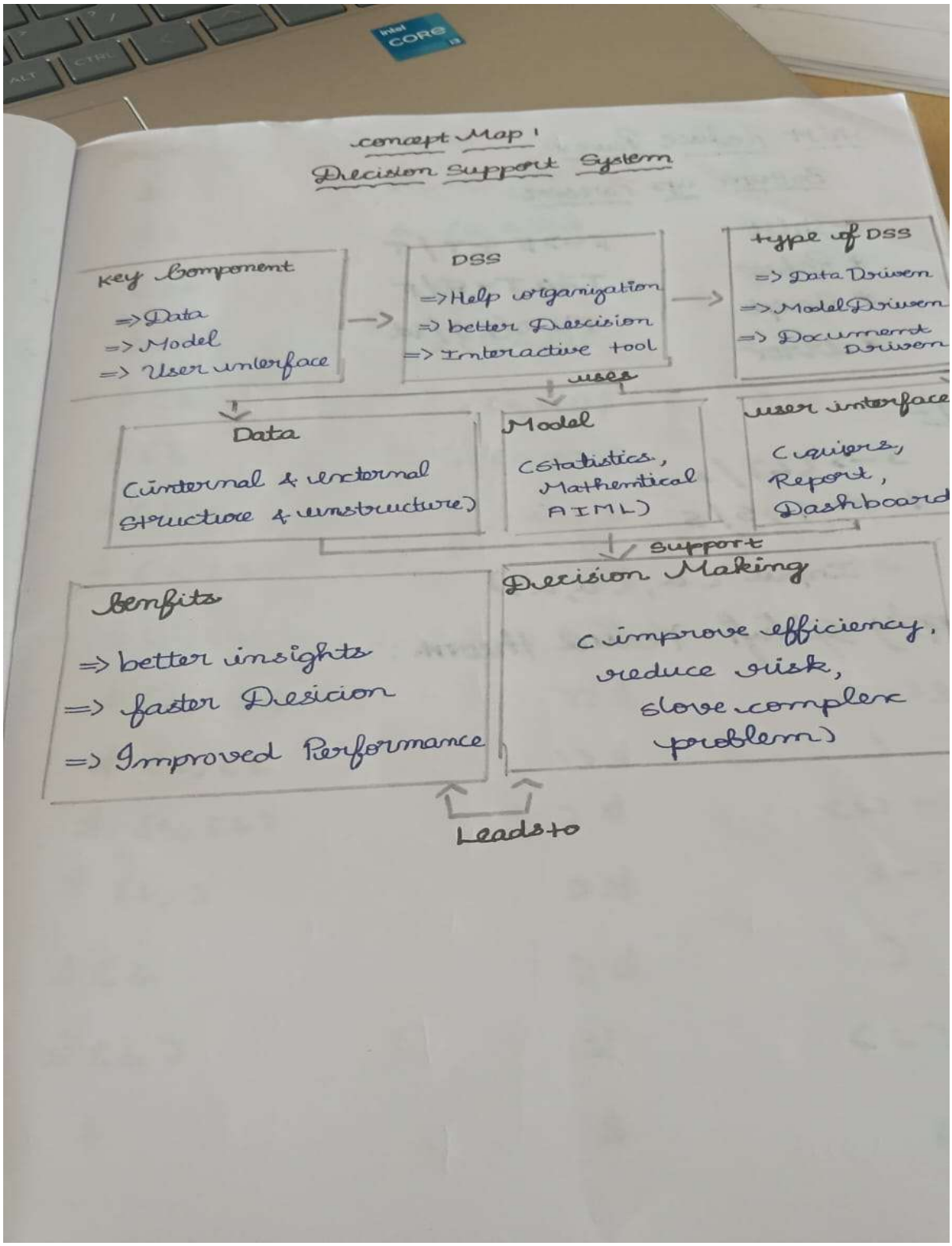
Task Description:

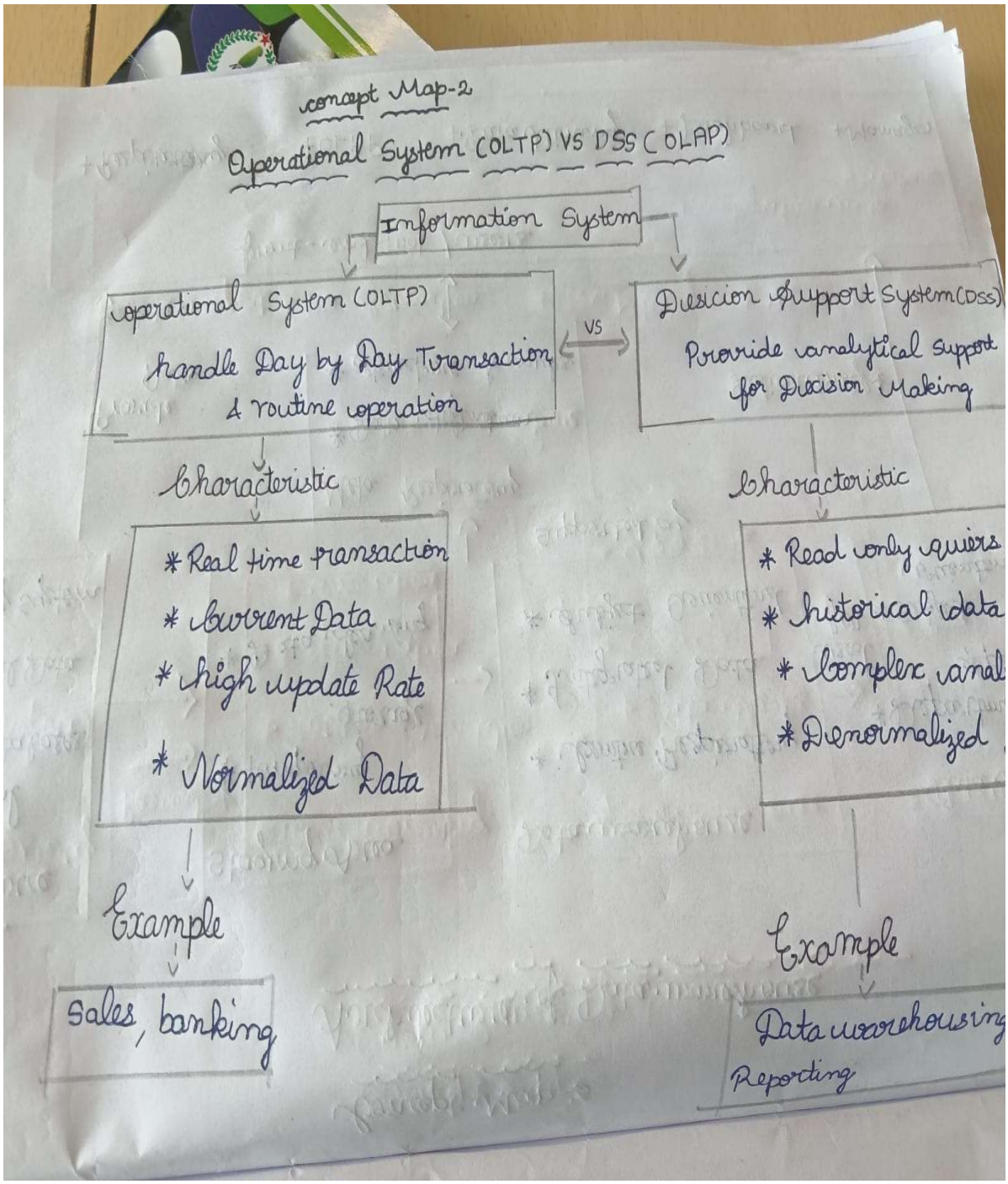
Construct a concept map explaining how query processing integrates with BI tools, focusing on KNIME for reporting and OLAP applications.

What your map must show:

1. Query processing stages (Parsing, Optimization, Execution)
2. Role of BI tools
3. KNIME workflow structure
4. Data → Query → Visualization pipeline
5. OLAP application integration

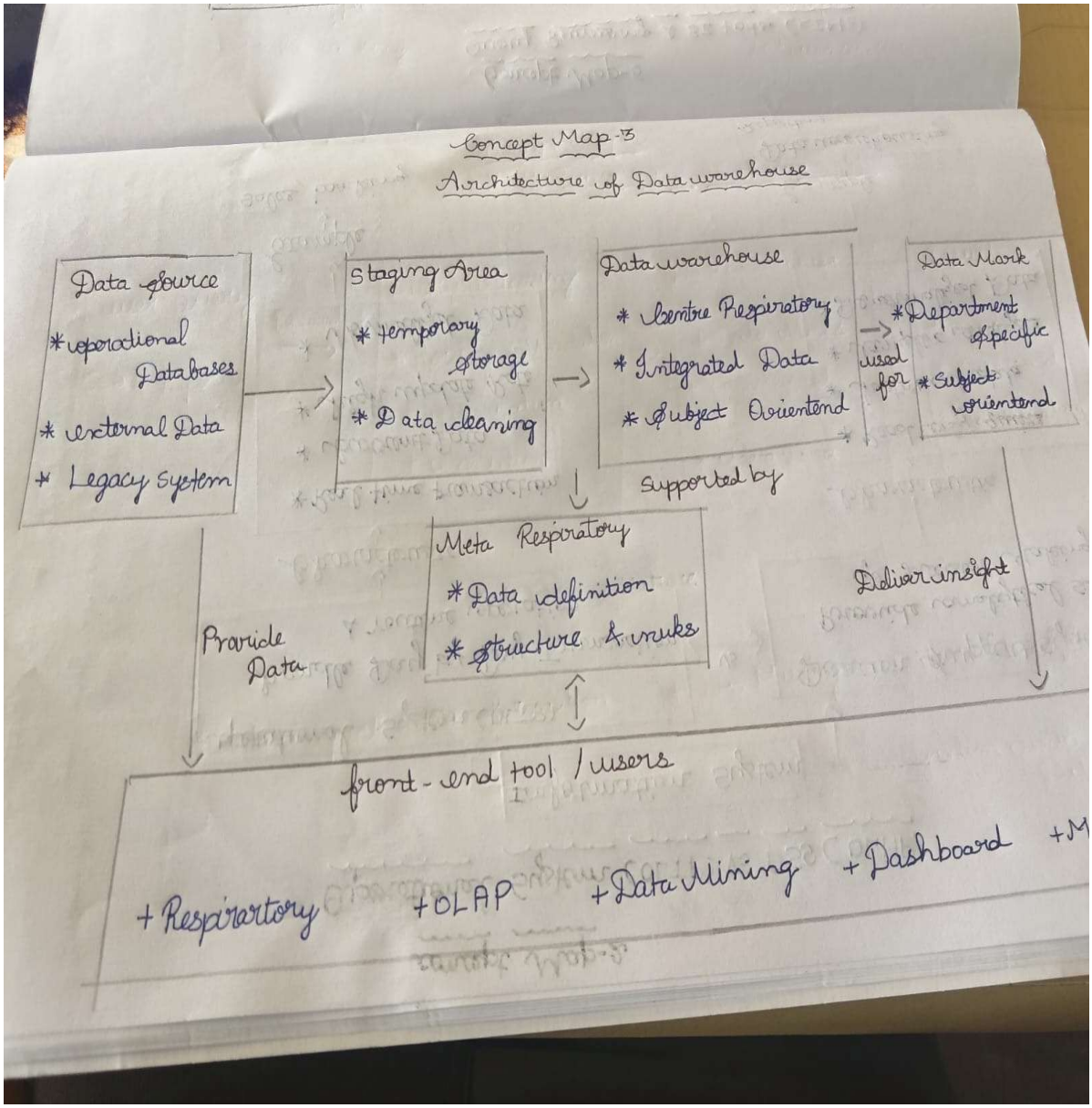
Answers

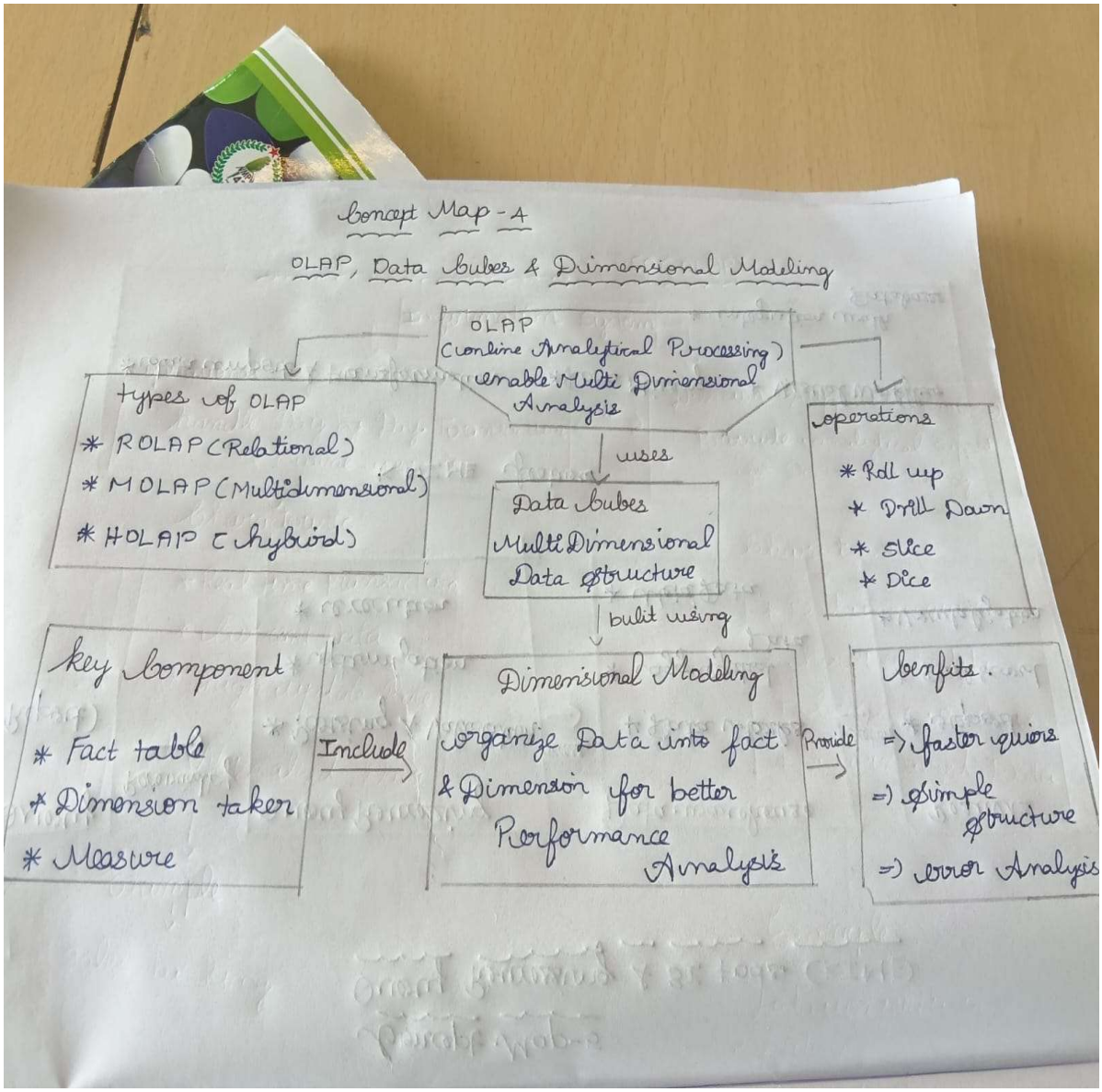

Concept Map :
Decision Support System
Key Component
=> Data
=> Model
=> User interface
DSS
=> Help organization
=> better Decision
=> Interactive tool
type of DSS
=> Data Driven
=> Model Driven
=> Document Driven
uses
Data
(internal & external
structure & unstructure)
Model
(Statistics,
Mathematical
AI ML)
user interface
(queries,
Report,
Dashboard
support
Decision Making
(improve efficiency,
reduce risk,
&love complex
problem)
benfits
=> better insights
=> faster Decision
=> Improved Performance
Leads to


concept Map-2
Operational System (OLTP) VS DSS (OLAP)
Information System
operational System (OLTP)
handle Day by Day Transaction & routine operation
VS
Decision Support System (DSS)
Provide analytical support for Decision Making
Characteristic
* Real time transaction
* Current Data
* High update Rate
* Normalized Data
Characteristic
* Read only queirs
* historical data
* complex anal
* Denormalized
Example
Sales, banking
Example
Data warehousing
Reporting

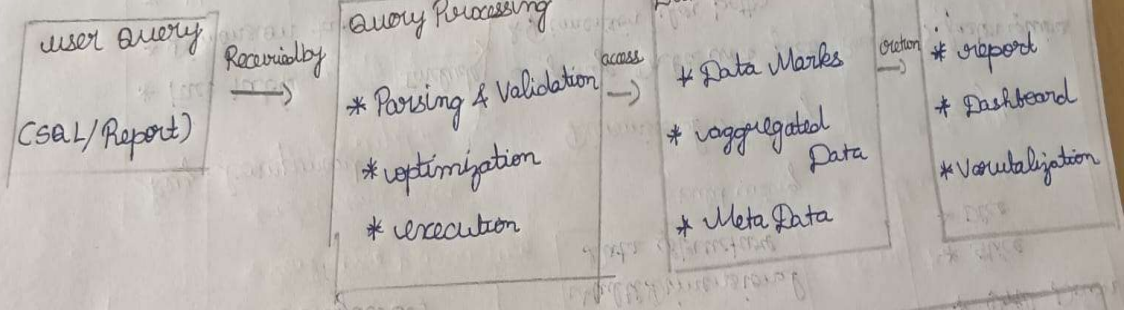
Concept Map-3

Architecture of Data warehouse




Concept Map - 4
OLAP, Data Cubes & Dimensional Modeling
OLAP
(Online Analytical Processing)
enable Multi Dimensional
Analysis
types of OLAP
* ROLAP (Relational)
* MOLAP (Multidimensional)
* HOLAP (hybrid)
uses
Data Cubes
Multi Dimensional
Data structure
built using
operations
* Roll up
* Drill Down
* Slice
* Dice
key component
* Fact table
* Dimension table
* Measure
Include
Dimensional Modeling
organize Data into fact
& Dimension for better
Performance
Analysis
Provide
benfits.
=> faster query
=) simple
structure
=) error Analysis

Concept Map-5
Query Processing & BI tools (KNIME)



KNIME feature

* data import & transformation
* Data Missing
* Visual workflow
* integration with Datab